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Food Chain with the 10% Energy Rule

An agent-based model of trophic collapse and apex-predator survival

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Agent Systems

The problem & our question

- **Lindeman's 10% rule:** each trophic level passes only ~**10%** of its energy to the level above.
- Five levels: **L1 grass** → **L2 herbivores** → **L3** → **L4** → **L5 apex predator**.
- Energy is lost at every step → the top of the pyramid is the most fragile.

Research question

Under the hard 10% rule, **how long does a 5-level chain persist**, and what determines the **survival of the apex predator (L5)**?

Our approach - the agents

- An **agent-based model** (not a cellular automaton): individuals move in **continuous space**.
- Each agent carries **energy** and follows simple per-tick rules:
 - **hunt / flee** (steering), **eat** → gain **+10% of the prey's energy**,
 - **reproduce** above a threshold, **starve** when energy hits zero.
- Producers (L1) photosynthesise and spread up to a carrying capacity.
- A single **deterministic tick loop** updates everyone; births/deaths applied atomically → no double-eating, reproducible per seed.

The simulator - live demo

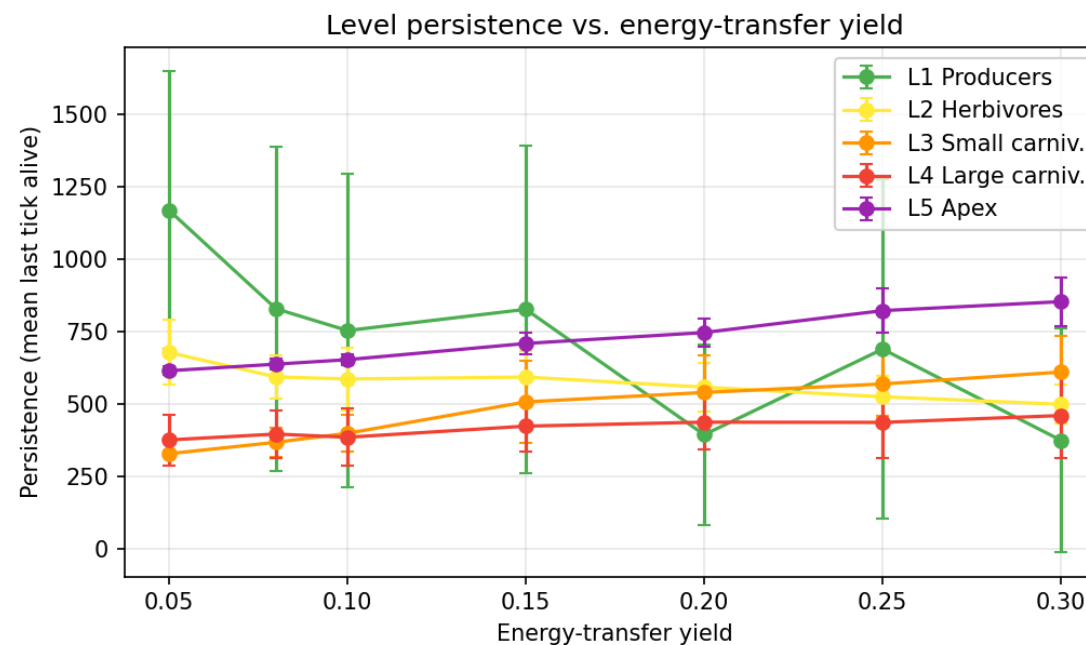


- Colours = trophic levels · brightness = energy · live **population chart** below.
- Sidebar = experiment knobs: **energy-transfer %**, metabolism, behaviours, seed, speed.

Method - measuring persistence

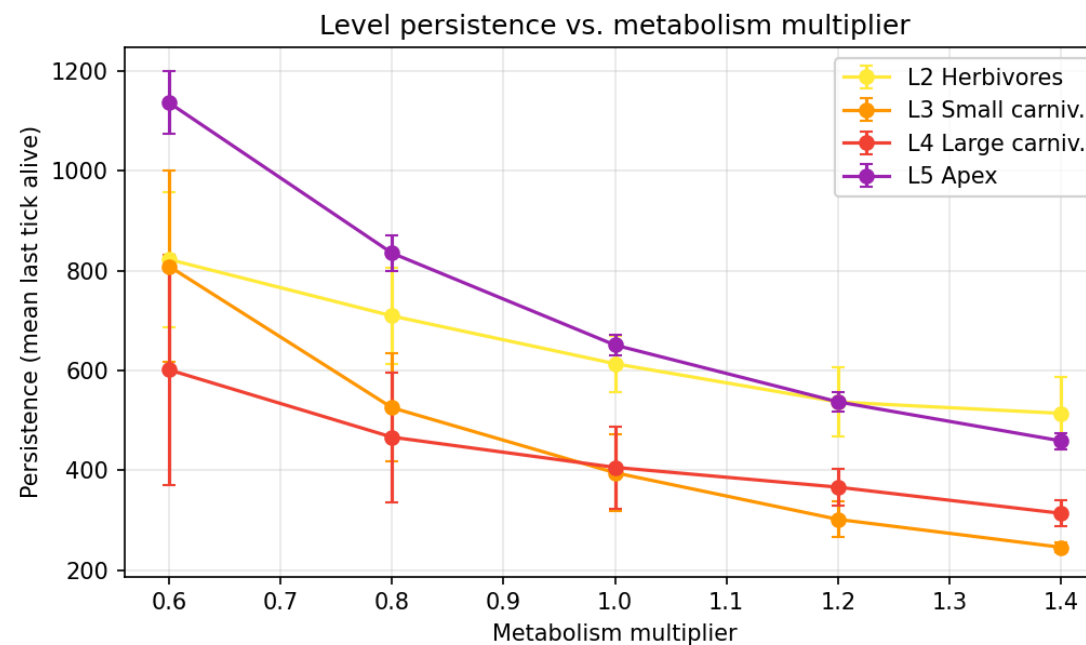
- **Metric - persistence:** the last tick at which a level still has ≥ 1 individual (longer = survived longer).
- **Headless parameter sweep:** same engine, no GUI; **30 random seeds**, reported as **mean $\pm$ SD**.
- **Why timing, not survival?** Without grass regrowth, *empty grass = an absorbing state* $\rightarrow$ every consumer level eventually collapses. So we measure **how long**, and show the **spread** across seeds.

Result 1 - the 10% bottleneck



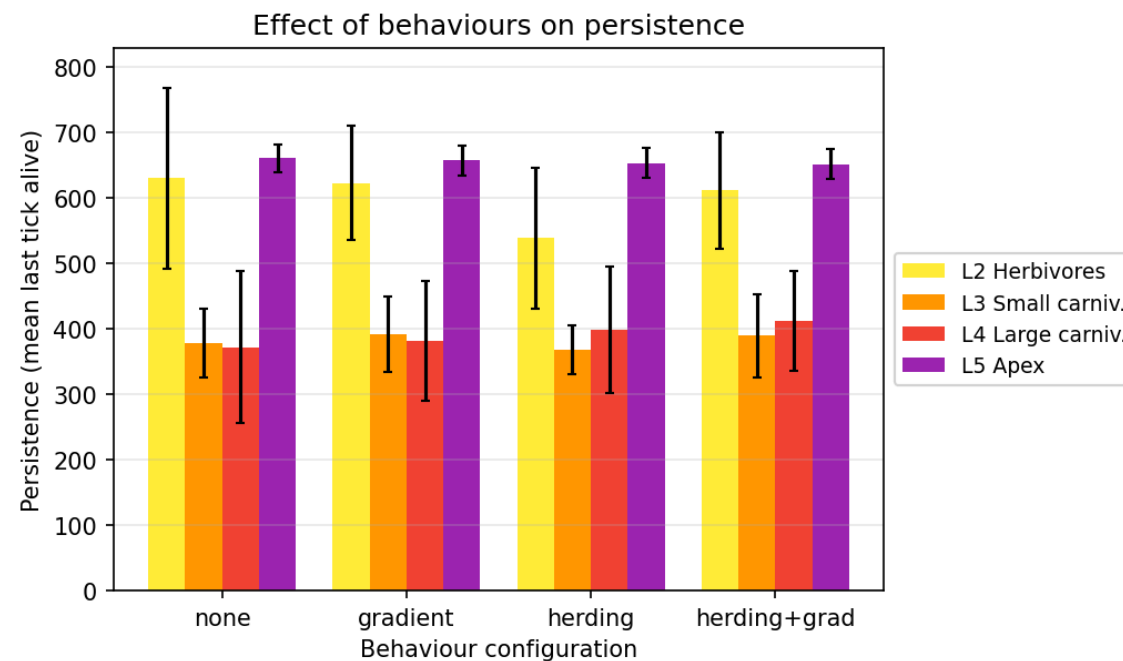
- Extinction order **L4 → L3 → L1 → L5 → L2** - the apex **starves last**.
- Higher yield → **longer apex persistence (614 → 853 ticks)**. · *error bars = ±1 SD*

Result 2 - what keeps the apex alive...



- **Strongest, most robust lever:** L5 lasts **1136** → **459** ticks ($\times 0.6 \rightarrow \times 1.4$).
- Monotonic; error bars don't overlap → a real effect.

...and what doesn't - behaviours



- All four configs overlap within ± 1 SD (L4 ≈ 372 – 412 , SD ≈ 77 – 117) $\rightarrow$ no real effect.
- Behaviours change **spatial structure**, not **survival timing**. (earlier "+28%" was a 15-seed artefact)

Conclusions & further work

Findings

- **Higher trophic level** → **lower survival probability** - the apex is the most fragile.
- The apex's fate is set by the **level below it**, not by its own behaviour.
- Strongest levers: **metabolism > energy-transfer yield**; behaviours = spatial structure only.

Further work

- Background grass spawn → recovery and sustained oscillations instead of one-way collapse.
- Trait inheritance + mutation → natural selection / ML directions.